# Practical No. 8

# SQLite CRUD – Contact App

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## Aim

To develop a simple Android application using **SQLite database** to store, retrieve, update, and delete contact information (name and phone number) and display the contacts using a ListView.

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## Software Requirements

• Android Studio
• Kotlin
• Android SDK

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## Step 1: Create a New Android Studio Project

• Language: Kotlin
• Template: Empty Views Activity
• Min SDK: API 21 (or default)
• File Name: contactapp

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## 📄 Step 2: Layout (activity_main.xml)

Very basic UI:
➡️ One EditText (Name)
➡️ One EditText (Phone)
➡️ One Button (Add)
➡️ One Button (View)
➡️ One Button (Update)
➡️ One Button (Delete)
➡️ One ListView

```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="10dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/name"
        android:hint="Enter Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/phone"
        android:hint="Enter Phone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/add"
        android:text="Add"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/view"
        android:text="View"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/update"
        android:text="Update"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/delete"
        android:text="Delete"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

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# Step 3: Kotlin Code (MainActivity.kt)

This is the entire logic. Simple and direct.

```
package com.example.contactapp

import android.database.sqlite.SQLiteDatabase
import android.os.Bundle
```

```
import android.widget.*
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    lateinit var db: SQLiteDatabase
    lateinit var listView: ListView

    var selectedId = -1

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val name = findViewById<EditText>(R.id.name)
        val phone = findViewById<EditText>(R.id.phone)

        val add = findViewById<Button>(R.id.add)
        val view = findViewById<Button>(R.id.view)
        val update = findViewById<Button>(R.id.update)
        val delete = findViewById<Button>(R.id.delete)

        listView = findViewById(R.id.listView)

        // create database
        db = openOrCreateDatabase("ContactsDB", MODE_PRIVATE, null)

        // create table
        db.execSQL(
            "CREATE TABLE IF NOT EXISTS Contacts(" +
                    "id INTEGER PRIMARY KEY AUTOINCREMENT, " +
                    "name TEXT, phone TEXT)"
        )

        // ADD
        add.setOnClickListener {

            db.execSQL(
                "INSERT INTO Contacts(name, phone)
VALUES('${name.text}','${phone.text}')"
            )

            Toast.makeText(this, "Contact Added",
Toast.LENGTH_SHORT).show()
        }

        // VIEW
        view.setOnClickListener {

            val cursor = db.rawQuery("SELECT * FROM Contacts", null)

            val list = ArrayList<String>()
            val ids = ArrayList<Int>()

            while (cursor.moveToNext()) {

                val id = cursor.getInt(0)
                val nameValue = cursor.getString(1)
                val phoneValue = cursor.getString(2)
```

```
                ids.add(id)
                list.add("$nameValue - $phoneValue")
            }

            cursor.close()

            val adapter = ArrayAdapter(
                this,
                android.R.layout.simple_list_item_1,
                list
            )

            listView.adapter = adapter

            // click contact
            listView.setOnItemClickListener { _, _, position, _ ->

                selectedId = ids[position]

                val parts = list[position].split(" - ")

                name.setText(parts[0])
                phone.setText(parts[1])

                Toast.makeText(this, "Contact Selected",
Toast.LENGTH_SHORT).show()
            }
        }

        // UPDATE
        update.setOnClickListener {

            if (selectedId != -1) {

                db.execSQL(
                    "UPDATE Contacts SET name='${name.text}',
phone='${phone.text}' WHERE id=$selectedId"
                )

                Toast.makeText(this, "Contact Updated",
Toast.LENGTH_SHORT).show()

            } else {

                Toast.makeText(this, "Select contact first",
Toast.LENGTH_SHORT).show()
            }
        }

        // DELETE
        delete.setOnClickListener {

            if (selectedId != -1) {

                db.execSQL(
                    "DELETE FROM Contacts WHERE id=$selectedId"
                )

                Toast.makeText(this, "Contact Deleted",
Toast.LENGTH_SHORT).show()
```

```
            } else {

                Toast.makeText(this, "Select contact first",
Toast.LENGTH_SHORT).show()
            }
        }
    }
}
```

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# Output

• User enters Name and Phone Number
• Click Add
• Contact is saved in SQLite database

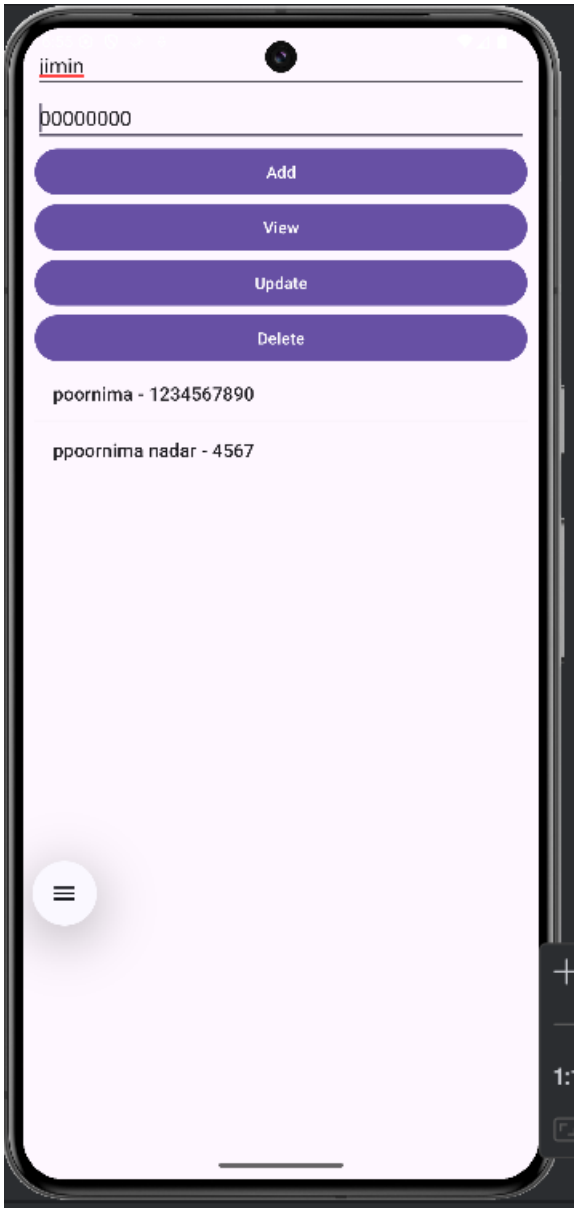
• Click View
• Saved contacts are displayed in ListView

• User selects a contact from ListView
• Selected contact details appear in input fields

• Click Update
• Selected contact is updated in SQLite database

• Click Delete
• Selected contact is deleted from SQLite database

• ListView displays the updated contact list accordingly


jimin
poooooooo
Add
View
Update
Delete
poornima - 1234567890
ppoornima nadar - 4567